

Junyoung Park

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RESEARCH INTERESTS

- Distributional data analysis, optimal transport; applications to wearable device data
- High-dimensional compositional data analysis; applications to microbiome data
- Non-Euclidean data analysis, sufficient dimension reduction, kernel methods

PROFESSIONAL POSITIONS

Postdoctoral Research Fellow

09/2024 –

Department of Biostatistics, University of Michigan, Ann Arbor, USA
Advisor: Irina Gaynanova

BK21 Postdoctoral Research Fellow

03/2024 – 08/2024

Natural Science Research Institute, KAIST, Daejeon, Korea
Advisor: Cheolwoo Park

EDUCATION

Ph.D. in Mathematical Sciences

03/2018 – 02/2024

KAIST, Daejeon, Korea
Thesis: “Kernel Methods for Compositional Data and Dimensionality Reduction”
Advisors: Cheolwoo Park, Jeongyoun Ahn

B.S. in Mathematics

03/2013 – 02/2018

Korea University, Seoul, Korea

AWARDS AND HONORS

Competitive Travel Awards and Invitations

- **IMS New Researchers Conference**, UMass Amherst, MA, USA 2026
- **IMS New Researchers Travel Award** 2026
- **Long Program on Connectomics**, Institute of Mathematical and Statistical Innovation (IMSI) 2026
- **1st IMS International New Researchers Conference**, University of Seville, Spain 2025

Awards and Scholarships in Korea

- **Best Ph.D. Dissertation Award**, Korean Statistical Society (2 awardees/year, prize $\approx$ \$2,000) 2025
- **Presentation Award for Graduate Students**, Korean Statistical Society (prize $\approx$ \$200) 2022
- **Outstanding Teaching Assistant Award**, Calculus II, KAIST Fall, 2020
- **University Students Contest of Mathematics**, Silver Awards, Korean Mathematical Society 2016, 2017
- **Presidential Science Undergraduate Fellowship**, fully funded for 8 semesters 2013–2017
- **The Korean Mathematical Olympiad (KMO) Gold Awards**, 2nd round, middle school division 2009

EXPERIENCE

Research Assistant in Statistical Learning

KAIST

Under Prof. Cheolwoo Park and Jeongyoun Ahn

08/2021 – 02/2024

– Kernel methods, compositional data, dimension reduction

Research Assistant in Algebraic Geometry

KAIST

Under Prof. Sijong Kwak

09/2018 – 07/2021

– Projective algebraic geometry, applied algebraic geometry

Technical Research Personnel

KAIST

For military obligation in South Korea

03/2020 – 02/2023

RESEARCH PAPERS

Underline denotes mentored Ph.D. student

Preprints/Submitted:

9. **Park, J.** and Gaynanova, I. (2026+) “Fréchet regression of multivariate distributions with nonparanormal transport.” Under review at *Journal of the Royal Statistical Society Series B: Statistical Methodology* (JRSSB). Preprint available on [arXiv:2603.07014](#).
– 2026 IMS New Researchers Travel Award
8. Shao, E., **Park, J.**, Punjabi, N., Jiang, H., and Gaynanova, I. (2026+) “Fast distance computation between multivariate distributions via nonparanormal Transport.” Under ‘Revise and Resubmit’ at *Journal of Computational and Graphical Statistics*. Preprint available on [arXiv:2603.00322](#). (Shao, E. received 2026 IMS Hannan Graduate Student Travel Award)
7. **Park, J.**, Park, C., and Ahn, J. (2026+) “Geometry-preserving and interpretable dimension reduction of compositional data.” Under review at *Journal of Machine Learning Research* (JMLR). Preprint available on [arXiv:2509.05563](#).

Peer-reviewed publications:

6. Xu, C., Schmidt, B.M., Krambrink, A., **Park, J.**, Song, P.X.-K., Jones, T.L.Z., Holmes, C., Chen, K., Ye, W., Kolenic, G., Roy, S., Sen, C., Pop-Busui, R., Spino, C., and Gaynanova, I. (2026) “Longitudinal Healing and Amputation Trajectories in Diabetic Foot Ulcers: Predictive Power of Wound Area and Duration and Sample-Size Implications From the Diabetic Foot Consortium.” *Wound repair and regeneration*, 34, No. 4, e70189. [Link](#).
5. **Park, J.**, Kok, N., and Gaynanova, I. (2026) “Beyond fixed thresholds: optimizing summaries of wearable device data via piecewise linearization of quantile functions.” *Statistics in Medicine*, Vol. 45, No. 15-17, e70646. [Link](#). R package [OptiThresholdR](#).
4. Yeon, K., Gao, F., Kim, R., Lee, D., **Park, J.**, Brown, D.A., and Park, C. (2026) “A Bayesian Approach to Semi-Supervised Domain Adaptation in Streaming Data.” *Journal of Statistical Computation and Simulation*, 96(9), 2237–2262. [Link](#).
3. **Park, J.**, Ahn, J., and Park, C. (2023), “Kernel Sufficient Dimension Reduction and Variable Selection for Compositional Data via Amalgamation.” *International Conference on Machine Learning* (ICML), pp. 27034–27047, PMLR. [Link](#).
2. Kang, I., Choi, H., Yoon, Y.-J., **Park, J.**, Kwon, S.-S., and Park, C. (2023), “Fréchet Distance-Based Cluster Analysis for Multi-Dimensional Functional Data.” *Statistics and Computing*, 33(4), 75. [Link](#).
1. **Park, J.**, Yoon, C., Park, C., and Ahn, J. (2022), “Kernel Methods for Radial Transformed Compositional Data with Many Zeros.” *International Conference on Machine Learning* (ICML), pp. 17458-17472, PMLR. [Link](#)

In preparation:

10. Multiple Imputation of Missing Predictors and Inference in Fréchet Regression of Distributional Responses (**Park, J.**, Szeto, N., Qian, L., Kok, N., and Gaynanova, I.).

TALKS

Invited talks:

- (Session is under review) **ENAR 2027 Spring Meeting** 2027
- (Scheduled) **IMSI Long Program**, during the invited visit to Connectomics program 2026
- (Scheduled) **International Conference on Computational Statistics (CompStat)** 2026
 - Session: *Advances in distributional data analysis*
- **IMS New Researchers Conference** 2026
- **1st IMS International New Researchers Conference** 2025
- **KAIST, Graduate School of Data Science** 2025
 - Title: Distributional data analysis approaches for wearable device data

Contributed talks/posters:

- **Mobile Technologies Research Innovation Collaborative (MeTRIC) Symposium** (poster) 2026
 - Beyond fixed thresholds: optimizing summaries of wearable device data
- **IMS International Conference on Statistics and Data Science (ICSIDS)** 2025
 - Fast Fréchet regression of multivariate distributions with application to wearable device data
- **Joint Statistical Meetings** 2025
 - Beyond fixed thresholds: optimizing summaries of wearable device data
- **Joint Statistical Meetings** 2024
 - Interpretable dimension reduction for compositional data
- **Winter Conference, the Korean Statistical Society** 2023
 - Interpretable composition-to-composition dimension reduction via conditional covariance operator
- **International Conference on Machine Learning (ICML)** (poster) 2023
 - Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **Summer Conference, the Korean Statistical Society** 2023
 - Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **International Conference on Machine Learning (ICML)** (spotlight talk and poster) 2022
 - Kernel methods for radial transformed compositional data with many zeros
- **2022 Summer Conference, the Korean Statistical Society** 2022
 - Kernel methods for radial transformed compositional data with many zeros (**presentation award**)

ACADEMIC SERVICE

- **Journal Refereeing**
 - Journal of Royal Statistical Society, Series B (1); Journal of American Statistical Association (1); Annals of Applied Statistics (2); Biometrics (3); WIREs Computational Statistics (1)

TEACHING

- **Teaching Assistant** at KAIST (selected list)
 - Statistical Data Science Practice (DS516) Spring 2023
 - Probability and Statistics (MAS250) Fall 2021
 - Mathematical Statistics (MAS355) Fall 2019
 - Matrix Group Theory (MAS435) Spring 2019
 - Gave a guest lecture on connectedness of Lie groups in English
 - Abstract Algebra II (MAS312) Fall 2018
 - Gave several guest lectures throughout the semester

COMPUTING SKILLS

- Python, R, Rcpp, R package development
- Optimization: autograd using `Pytorch`, just-in-time compiler (`numba`)
- Linux environment, high-performance computing cluster (slurm) experiences

LANGUAGES

- English (fluent)
- Korean (native)

HOBBIES

- Singing, better with playing guitar
- Coffee